HAT-P-32b

R-0258

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-32_220920` · created 2026-07-17T08:17:33+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459842.7614 +/- 0.0038 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.047
Transit depth (Rp/Rs)^2	5e-06 +/- 0.0021 [%]
Orbital Inclination (inc)	89.19 +/- 1.72
Airmass coefficient 1 (a1)	4.66 +/- 2.71
Airmass coefficient 2 (a2)	-1.37 +/- 0.18
Scatter in the residuals of the lightcurve fit is	43.55 %
Best Comparison Star	None
Optimal Aperture	2.52
Optimal Annulus	4.18
Transit Duration (day)	0.116 +/- 0.0034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.0 ± 0.047 vs 0.1489 (deviation 1.52σ)
Duration fit vs published	0.116 d vs 0.1304 d (deviation 2.04σ)
Mid-time O–C	-8.9 min
Depth SNR	0.0
Residual scatter	43.55 %

Light curve

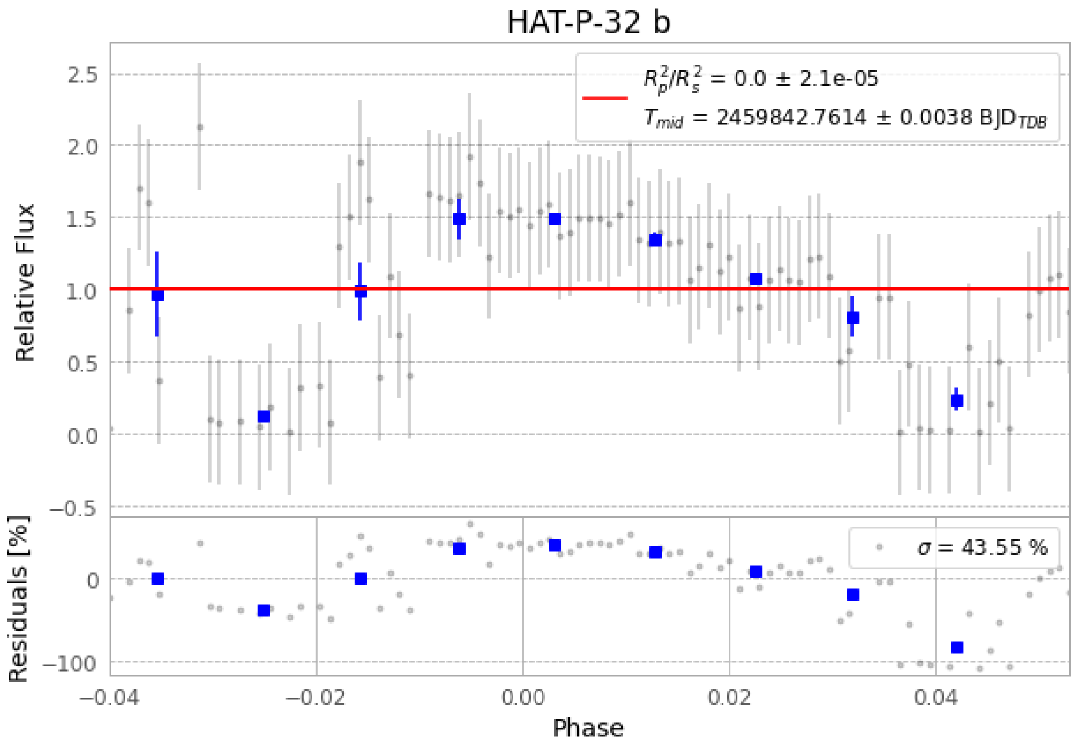

Figure 1 — FinalLightCurve_HAT-P-32 b_2022-09-19.png (SHA-256 549f3b2d44f18d11…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [337, 77], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 c6aa226c3f1b0c98…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit cbb899b`

Data provenance

97 FITS frames (64 MB), `HATP-32220920040615.FITS` → `HATP-32220920085415.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-220920.log (3bcff651e4b2…), FinalLightCurve_HAT-P-32 b_2022-09-19.png (549f3b2d44f1…), FinalLightCurve_HAT-P-32 b_2022-09-19.csv (a23bc5d9eb8d…)

Compute resources

Wall time 250.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)